01/12 Thermo syllabus

Monday, 12 January 2026 10:30 AM

Final project → Catalysis (Adsorption mechanism, kMC)

→ Capital → extensive } Convention.
letters (depend on the amount of substance) }

→ Postulate I (Duhem's) → n species; n+2 variables for bulk properties

→ Macro vs. Micro

↳ average behavior of length & time scales

Kinetic Theory of Gases

- Dilute (i.g. limit)
- non-interacting hard spheres
- Neutral atoms
- Polarizability of atom (induced dipole)

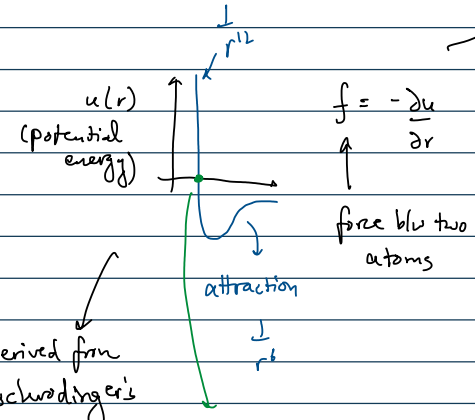
of molecules ↳ $N = nN_A$ (n: # of mols)

V → extensive volume

$\rho = \frac{N}{V}$ → no. density

$k = \frac{R}{N_A} = 1.38 e^{-23}$ J/K

$$\boxed{\rho = \frac{N}{V} = \frac{P}{kT}}$$



derived from schrodinger's eq

$f = -\frac{\partial u}{\partial r}$ ↑ force b/w two atoms

Coulomb's law still at play but these are averaged (so no two particles)

0 point ($u(r)=0$) gives $r=\sigma$ as effective size

6-12 Leonard Jones } very good model for noble gases (Xe, Ar, Kr)

gas → 10σ separation

liq → 1σ separation

01/14

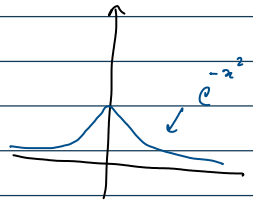
Wednesday, 14 January 2026 10:31 AM

Assumptions : + elastic collisions
+ no external forces ; so density (ρ) independent of position
+ KE is distributed independently and in an equal manner } principle of equipartition
↳ $h(r+s+t) = g(r)\, g(s)\, g(t)$ KE scaling as v^2 or p^2

Maxwell distribution : $g(v_x^2) = \alpha\, e^{-a v_x^2}$

↳ probability

$$\int_{-\infty}^{\infty} f_x(v_x^2) \cdot dv_x = 1$$



Derivations → $|f(\vec{v})| = f(|v|) = f(v^2)$

$\Phi = \frac{\Delta N}{\Delta A\, \Delta t}$ → dependent on probability of particles of v going at an angle

molecular flux

↳ @ equilibrium

$F = \frac{dp}{dt}$ ← change in momentum

kT → energy scale of average KE, or motion of a particle

01/16

Friday, 16 January 2026 10:35 AM

HW due 02/06 → 6 questions (last question is coding)

→ Average pressure exerted on the walls of the container ⇒ $P = \frac{3}{2} k_B \rho T$; $\bar{\varepsilon} = \frac{3}{2} kT$

→ Experiments done confirming the maxwell-boltzmann distribution of speeds.

↳ starts to fail in quantum scale.

→ Q → Why is there $N_2 > H_2$ in the atmosphere?

↳ difference in the average speed of the molecules as function of temp. (enough to leave the atmosphere due to their higher escape velocity)

01/21

Wednesday, 21 January 2026 10:30 AM

In a liquid, molecules are "always colliding"; gases have a mean free path (distance b/w each collision)

Probability of collisions + No. of particles incident for a cross sectional area → gives survival equation

$l = \frac{1}{\pi \sigma^2 n}$

Effusion: gas leaks through a very small hole of diameter much smaller than the mean free path

$d \ll l$, molecules will undergo very few collisions

↳ this is where the "PV" term comes in idea of viscous stresses when forcing gas/liquid through orifice

Effusion is when the collision has already occurred & the gas just passes through.

Viscous stresses arise from net transport of momentum via gases moving at different speeds & colliding w/ other particles/molecules.

Week, 26th Jan

Monday, 26 January 2026 10:32 AM

Lagrangian functions to define <u>microstates of system</u>

↳ radians & sec. (distance & time)

q → velocity $(\dot{q}_i)$ $\quad i = 1, \ldots, 3N$

L → units of energy

$L = KE - PE$

$$KE = \frac{1}{2} m_i \dot{q}_i^2$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = m_i \dot{q}_i$$

↑ momentum

Hamiltonian is legendre transform of L (Lagrangian)

$H = KE + PE$ } total energy of the system

Phase space → q & p space

↑ momenta

→ Conservative system (energy conserved)

→ Isolated system (N, V,)

Macro —statistical mech.→ Micro

(6N dof)

↓

Experiments ↘

Data is here

Micro-canonical → divide system is smaller ensembles
& have initial conditions

↳ & instead of projecting / solving Lan. & equations for each particle]

use stats & subject collections from ensembles.

Q = # of members of the ensemble

f = phase space density (p, q)

→ Hamilton's equation of Motion

→ Liouville's Theorem

Getting rid of the dynamic evolution but sampling states at fixed times

↓

States w/ same energy will be sampled w/ the same frequency.

Postulate of equal a-priori probability

+ isolated, thermal equilibrium

+ time-independent ensemble average $\langle F \rangle$

→ Ergodic: time averages equivalent to just counting states
hypothesis (don't have to integrate any equations of motion)

All possible p, q that have the same E, are sampled w/ the same frequency

→ Global properties

Lagrange multiplier

P_ν → probability of microstates

Ω → # of microstates w/ given N, V, E

Fundamental eq.: $dU = TdS - pdV + \mu dV$

Boltzmann's Entropy formula ⇒ $\boxed{S = k \ln \Omega}$

→ Allowed microstates can have fluctuated volume

$\Delta S < 0$ → equally likely, and equal sample frequency

$\Omega(v) = v^N$

$\Omega(V) = V^N$

Enormous # of microstates even for a low entropy system

fluctuations → But the overall probability of this

fluctuations allow for this → But the probability of this happening is exceedingly small

(In the thermal limit)

> Entropy is a measure of accessible microstates

Week, 2nd Feb

Monday, 2 February 2026 10:32 AM

→ Simple model of elasticity

n_+ & n_- → orientation of monomer

$\Omega_{chain} = \frac{N!}{n_+! n_-!} = \frac{N!}{n_+!(N-n_+)!}$

→ Fundamental equation for elastic band

$dU = TdS + \tau\, dL + \mu dN$

tension ↓

$\tau \Leftrightarrow -p$

↑ acts opp. to pressure

$L \Leftrightarrow x$

→ As stretching out, the entropy goes down ↓

only limited arrangements for the microstates of monomers

$\left.\frac{\partial T}{\partial x}\right|_{S,N} > 0$

$\Omega(N,V,E) = \int_{E \le H \le E+\delta E} d\underline{p}\, d\underline{q}$ } is a volume in phase space in classical mechanics.

(Not dimensionless in this current eq.)

$\Delta p \Delta q \gtrsim \frac{h}{4\pi}$; $H = KE + pE$

p: position

Classical mechanics overcounts the microstates; but certain distances (less than h) cannot be measured.

for ideal gas: $\Omega(N,V,E) = V^N \int \cdots \int dp \cdots dp$

for ideal gas: $\Omega(N,V,E) = \frac{V^N}{h^{3N} N!} \int \cdots \int dp_1 \cdots dp_{3N}$

$0 \leq \frac{p^2}{2m} \leq E$

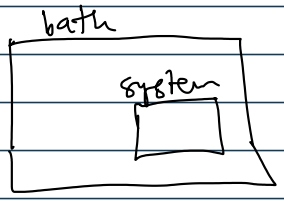
Gamma function,

Volume of n-dimensional hyperspace $\rightarrow V_N =$

Just from Hamiltonian & microstates $\rightarrow$ we derive everything

Canonical Ensemble (N, V, T) (Microstates vs H, V, E)

(Allows for energy exchange)



HW#1

Friday, 6 February 2026 10:53 AM

Q1] Effusion of gas through a small hole

$f(\vec{v})\,d\vec{v}$ = probability of a given molecule having a velocity vector b/w $\vec{v}$ and $\vec{v}+d\vec{v}$

$$f(\vec{v}) = f(|\vec{v}|) = f(v^2) \quad ; \quad \iiint f(\vec{v})\,d^3v = 1$$

$$f(v^2) = C\,e^{-av^2} \quad ; \quad a>0$$

(Maxwell Distribution)

$$\langle v_x \rangle = \bar{v}_x = \int_{-\infty}^{\infty} v_x f_x(v_x^2)\,dv_x$$

$$\langle v_x^2 \rangle = \overline{v_x^2} = \int_{-\infty}^{\infty} v_x^2 f_x(v_x^2)\,dv_x$$

$$\int_{-\infty}^{\infty} f(v^2)\,dv = 1$$

Solving this in spherical coordinates gives

$$\langle v \rangle = \bar{v} = 4\pi \int_0^{\infty} v\, f(v^2)\, v^2\,dv$$

$$\langle v^2 \rangle = \bar{v}^2 = 4\pi \int_0^{\infty}$$

Number of molecules crossing the hole per unit time w/ speed v is $\bar{v} f_E(\bar{v})\,d\bar{v}$

$$\langle v_E \rangle = \frac{\int_0^{\infty} \bar{v}\, f_E(\bar{v})\,d\bar{v}}{\int_0^{\infty} f_E(\bar{v})\,d\bar{v}}$$

↓

solving this integral using $f_E(\bar{v})\,d\bar{v} = \left(\frac{m}{2\pi kT}\right)^{1/2} e^{-m\bar{v}^2/2kT}\,d\bar{v}$ from Maxwell-Boltzmann dist.

& some math...

Similarly, $\langle v_E^2 \rangle = \int_0^{\infty} \bar{v}^2 f_E(\bar{v})\,dv$ this should yield $2kT$

Similarly, $\langle v_E^2 \rangle = \dfrac{\int_0^\infty \bar{v}^2 f_E(\bar{v})\, d\bar{v}}{\int_0^\infty f_E(\bar{v})\, d\bar{v}}$ → this should yield $2kT$

Q2] Maxwell-Boltzmann for a 2D gas ; v_x and v_y direction

Probability of speed for 2D ideal gas b/w $\bar{v}$ and $\bar{v}+d\bar{v}$ → $f_{2D}(\bar{v})\, d^2\bar{v} = \int_0^{2\pi} f(v_x, v_y)\, d^2v$

From Boltzmann dist., $f(v_x, v_y) = \dfrac{m}{2\pi kT} e^{-m(v_x^2+v_y^2)/2kT}$

$\bar{v}^2 = v_x^2 + v_y^2$ (Ideal gas)

Q6] Random walk in 3D

N → total number of steps during the walk

(a) $\langle R \rangle$, $\langle R^2 \rangle$ vs. N

↑ Average length

(b) $P(R) = 4\pi R^2 \left(\dfrac{3}{2\pi N}\right)^{3/2} \exp\left(\dfrac{-3R^2}{2N}\right)$

↳ probability density that the random walk has an end-to-end distance equal to R

What happens as $N \to$ large

Week of 9th Feb

Monday, 9 February 2026 10:30 AM

→ Molecular partition functions

Some choice of p & q at a certain energy

Sum of microstates individually for non-interacting particles

$$q = \sum_{\nu} e^{-\beta \nu E}$$

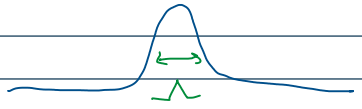
→ Third law of Thermodynamics

$T \to 0, \ \beta \to \infty$

Momenta can be split w/ position for integration.

$$Q = \frac{q^N}{N!} = \frac{V^N}{N! \Lambda^{3N}}$$

Canonical vs. Micro-canonical ??



→ Energy levels are decoupled

$$H = H_{trans} + H_{elec} + H_{rvib}$$

This can be assumed because the scale of things is very different. → Energy levels are ~ 10 eV while translation can occur & smaller eV's

Λ_i → de-Broglie wavelength

$$\Lambda_i = \left(\frac{h^2}{2\pi m_i kT} \right)^{1/2}$$

→ Jacobian does not apply to phase space transformations

$$p = kT \left. \frac{\partial \ln Q}{\partial V} \right|_{N,T}$$

$$Q = \sum_{\nu} e^{-\beta E_\nu} \qquad P_\nu = \frac{e^{-\beta E_\nu}}{Q}$$

$$U = \langle E_\nu \rangle = \sum_{\nu} P_\nu E_\nu = \frac{\sum_{\nu} E_\nu e^{-\beta E_\nu}}{} = \left. \frac{-\partial \ln Q}{\partial \beta} \right|$$

$$U = \langle E_V \rangle = \sum_V p_V E_V = \frac{\sum_V E_V e^{-\beta E_V}}{Q} = \left. \frac{-\partial \ln Q}{\partial \beta} \right|_{N,V}$$

C_V tells us the size of energy fluctuations of a system $\quad \langle (\delta E_V)^2 \rangle = kT^2 C_V$

At critical points the fluctuations in energy become meaningful

↳ $C_V \to \infty$ $\quad dV = C_V dT$

The states are being sampled extensively

where V population of states

temp. is used to vary the

→ Maxwell's demon

Schrodinger's Book → What is life?

Why are atoms so small?

→ Equipartition Theorem

Rotation Partition Function

$$q_{rot} = \frac{I}{\theta \sigma_{rot}}$$ (Classical limit for a diatomic)

Rotational temperature → ?

Rotational temperature → ??

Moment of inertia

↳ Finding degrees of freedom ; Symmetry no.

Potential Energy Surface

→ Multidimensional surface

→ @ const T, Boltzmann factor ($-\beta E$) indicates likely configurations

$\frac{kT}{h}$ → partition function for translation for the activated complex

Classical mechanics } · Know precise knowledge of positions, velocities, momentum
· Model the future evolution of system

Macro thermo } $N \sim 10^{23}$ particles
$6N$ degrees of freedom

We usually have information about the "macroscopic state" of system (V, E, T, etc.)
↳ How do we get the micro-state i.e. particular values of all $6N$ dof's.

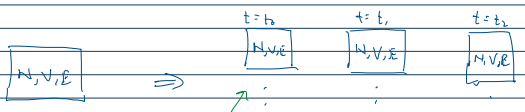
↓

Construct an ensemble of system → average behavior of a collection of identical N particle system distributed over a range of different possible states

They are not smaller systems or control volume

↓

More so representing the entire system & its time evolution w/ identical systems at different times (or initial conditions)



$\Omega = \sum$ ensembles (total ensembles)

TLDR:

- Each ensemble can be represented by a single point in $6N$ dimensional space ($3N$ position & $3N$ momentum)
- These ensembles are clubbed together in a phase space density ?

  Thus this cloud/collection of ensembles can be modelled like an incompressible flow

- This way we can use ensemble averages since if you wait long enough, a single system will visit every part of the phase space allowed by its energy

  ↓

  In a way the time evolution is just modelled as taking place already to give all the possible states → then you are just sampling from it.

  (This is the ergodic hypothesis → Boltzmann 1871)

Each ensemble has N particles whose dynamic behavior is given by Hamilton's eqn's of motion

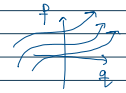
These are a set of first-order differential equations that describe how a system evolves through phase space (q_i, p_i)

H = total energy of system

$$\dot{q}_i = \frac{\partial H_i}{\partial p_i} \qquad \dot{p}_i = -\frac{\partial H_i}{\partial q_i}$$

↓

A point in phase space represents the complete state of system (position & momentum) → as time progresses this point traces a path called a trajectory



$f(p, q, t)$ = density in phase space

$\Omega = \int_{E \le \mathcal{H} \le E+\delta E} dp\,dq$ → volume of phase space whose energy lies b/w E and $E + \delta E$

↳ just a pure number quantum mechanically → no. of microstates for a given E

$$S = -k \sum_\nu P_\nu \ln(P_\nu)$$

(N, V, E) → constant

Micro canonical Ensemble

+ Describes an isolated system (no exchange w/ environment)

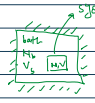
+ Principle of equal a-priori probability → every accessible microstate Ω is equally likely to be occupied

$$P_i = \begin{cases} \frac{1}{\Omega} & E_i = E \\ 0 & \text{else} \end{cases}$$

$$S = k_B \ln \Omega(N, V, E)$$

(energy can change)
(N, V, T) → constant

Canonical Ensemble



+ All microstates are not equally likely (since E is not constant) Probability of finding system in a microstate i w/ energy E_i given by the Boltzmann distribution

} specifically lower energies are favored $E_\nu \downarrow \rightarrow E_b \uparrow$

+ Probability of finding system in any microstate (ν) :

$$P_\nu = \frac{\Omega_b(E_b)}{\Omega(E)}, \quad E_b - E_\nu = E$$

↓ math & Taylor expansion

$$P_\nu = \frac{e^{-\beta E_\nu}}{\sum_\nu e^{-\beta E_\nu}}$$

} This is partition function (Q or Z)
+ Sum of the probabilities of all possible microstates equals 1
+ Has information about all the system's energy levels

} This is later used to bridge between micro and macro

$$A = -kT \ln Q$$

↓

For non-interacting independent particles the hamiltonian and hence the Energy microstates are separable

In the thermodynamic limit, both ensembles become equivalent

$$A = -kT \ln Q$$

$$U = \langle E \rangle = \left. \frac{-\partial \ln Q}{\partial \beta} \right|_{N,V} = \left. kT^2 \frac{\partial \ln Q}{\partial T} \right|_{N,V}$$

$$P = \left. k_B T \frac{\partial \ln Z}{\partial V} \right|_{N,T}$$

$$S = \frac{U - A}{T} = \left. k_B \ln Q + k_B T \frac{\partial \ln Q}{\partial T} \right|_{N,V}$$

For non-interacting, independent particles the Hamiltonian and hence the Energy microstates are separable

↓

Therefore we can assign

$$Q = \sum_\nu e^{-\beta E_\nu} = q_a q_b q_c \cdots$$

where $q_i = \sum_{\nu_i} e^{-\beta \epsilon_{\nu_i}}$ → molecular partition function

$q_i = q_i(V,T)$

Ideal gas

$$H = H_{trans} + H_{elec} + H_{vib}$$

quantum correction

$$Q = \frac{1}{h^{3N} N!} \int e^{-\beta H} \, dp \, dq$$